Rule: distribute dot over add

```
(dot[@d] (add A B) C)
→ (add (dot[auto] A C) (dot[auto] B C))
```

Reusable operation semantics

```
add: Add(X,Y)
dot: Reduce(K,Product(X,Y))
```

1. Axes IR

Source

```
Reduce(K,
  Product(Add(A,B),C))
```

Contract K; output I ++ J.

RHS template

```
Add(
  Reduce(K₁,Product(A,C)),
  Reduce(K₂,Product(B,C)))
```

K_1, K_2 unresolved

Reads retain coordinates.

2. Normalized terms

Source

```
[(1,[A,C],K),
 (1,[B,C],K)]
```

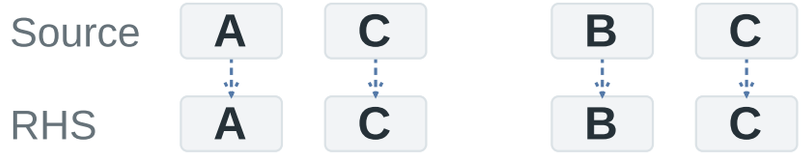
RHS arithmetic skeleton

```
[(1,[A,C],?),
 (1,[B,C],?)]
```

Term = (weight, reads, axes)
P_T: ordered axes of tensor T.
?: unresolved axis information

3. Attribute matching

Match operand reads



Reuse source coordinates

```
A,B: [I,K]; C: [K,J]
```

Construct attributes

Contract K; preserve I ++ J.

```
A → positions(P_A,K)
B → positions(P_B,K)
C → positions(P_C,K)
```

Constructed replacement

```
(add (dot[@d₁] A C) (dot[@d₂] B C))
```

$@d_1$ = @d; $@d_2$ = @d Independent holes

Verify completed rule

Legality, values, output order
No broadcast; distribution allowed.